

SILAJEET BANERJEE

MSc in Computer Science, RKMVERI, Belur

9007732623
silajeetbanerjee100@gmail.com
GitHub
LinkedIn
Portfolio



SUMMARY

Computer Science graduate with interests in Artificial Intelligence, Machine Learning, and intelligent systems. Skilled in software development, machine learning, and intelligent systems with experience building AI-driven applications and autonomous platforms. Seeking opportunities to apply analytical and research skills in Artificial Intelligence, Data Science, and intelligent systems.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Computer Science, 2025–27
CGPA – 7.17(till Sem 1)

University of Calcutta, Kolkata
BSc (Honours) in Computer Science, 2021–24
CGPA – 7.039

SKILLS

- **Languages:** C, C++, Java, Python, SQL
- **Frameworks & Libraries:** PyTorch, OpenCV, Scikit-Learn, NumPy, Pandas, Streamlit, Matplotlib, LangChain, LangGraph, AutoGen, crewAI
- **Operating Systems:** Windows, Linux
- **Tools:** Git, LaTeX, Jupyter Notebooks

Research Experience

ACM IKDD Uplink Summer Research Intern

May 2026 – Present

IIT Bombay, Mumbai, India

Supervisor: **Prof. Harshad Khadilkar**

- Designing and implementing a scalable, privacy-preserving, modular agentic pipeline for extracting and identifying research papers from Core A* venues with Indian-affiliated authors.

COURSE WORK

- Machine Learning, Deep Learning, Artificial Intelligence
- Probability, Linear Algebra, Mining of Massive Datasets
- Design and Analysis of Algorithms, Advanced Algorithms

ACHIEVEMENTS

- CUET-PG 2025 (Computer Science) – Qualified
- Awarded the Swami Vivekananda Merit Cum Means Scholarship
- Presented my project "Beyond the Hub: Exploiting Structural Bridges for Targeted Epidemic Control via Graph Reinforcement Learning" at the Ajit Diwan Memorial Workshop (2026).

PROJECTS

AEGISv2: An Agent for Extraction and Geographic Identification in Scholarly Proceedings

IIT Bombay

May 2026 – Present

Architected a custom local web-agent pipeline replacing Agent-E, reducing required LLM scale from $\geq 70B$ to $20B$ parameters and enabling 100% local execution on Apple Silicon with zero recurring cloud API costs. Scaled geographic scholarly extraction from 3–4 conferences to 10,000+ papers across 23+ CORE A* venues, automatically scraping and parsing HTML/PDFs and identifying Indian-affiliated authors. Integrated OpenWebUI for interactive natural language task execution and automated backend duplicate verification and submission to the ACM IKDD Premier Papers portal.

A.P.R.I.L.

Independent Open Source Project

May 2026 – Present

Engineered a local cognitive voice assistant using Whisper STT, Kokoro TTS, and local LLMs for real-time speech interaction. Architected a dual-stage memory substrate combining a sliding-window working context with a local Neo4j graph database for long-term semantic persistence. Built an MCP server ecosystem with deterministic system guardrails to decouple tool routing, enabling multi-hop entity resolution without conversational drift.

NetGuard AI – Rumor Containment via Graph Reinforcement Learning

RKMVERI

Sep 2025 – Dec 2025

Engineered a custom environment to simulate stochastic information diffusion (SIR dynamics) on modular social networks (Caveman graphs). Developed and compared MCTS, DQN, and GraphSAGE agents, designed an aggressive reward shaping mechanism to penalize total infection duration. Selected for presentation at the Ajit Diwan Memorial Workshop (January 2026), recognized for novel application of GNNs in viral containment.

StockLLM

RKMVERI

Feb 2026 – Apr 2026

Developed an AI-driven stock market forecasting system combining iTransformer and Temporal Fusion Transformer (TFT) models with technical indicators and live news sentiment analysis. Generated 5-day price forecasts and consensus BUY/SELL/HOLD recommendations using an LLM-powered decision orchestration pipeline.

NeuroGraph-X – Hybrid CNN-GNN for Alzheimer's Detection

RKMVERI

Feb 2026 – Apr 2026

Developed NeuroGraph-X, a hybrid CNN-GNN framework for early Alzheimer's progression prediction (pMCI vs sMCI) using structural MRI data from the ADNI dataset. Designed dual spatial and morphological brain graphs with edge-aware GATv2 attention and cross-attention fusion, achieving 73% recall for progressive MCI detection.

AI-Assisted Autonomous Ground Vehicle (AGV)

University of Calcutta

Apr 2024 – Jul 2024

Developed an AI-assisted autonomous robotic platform integrating YOLOv5-based human detection, real-time target tracking, and wireless control using ESP32, ESP8266, and Arduino microcontrollers. Implemented computer vision-guided servo actuation and remote monitoring for autonomous surveillance and target localization.